MODULE 2 · STATISTICS & PROBABILITY · TOPIC 2.4

The Normal Distribution

Day 24 — when the standard deviation becomes a ruler



Boston University

68-95-99.7	1.96	30 min
The rule worth memorising	The number behind every CI	Practice notebook

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WARM-UP • 5 MINUTES

Yesterday the SD described. Today it measures.

1	**What does n − 1 correct for?**	→ The sample mean sits too close to its own data.
2	**ddof for a whole population?**	→ ddof=0. Nothing is unseen.
3	**Why quote the SD, not the variance?**	→ The variance is in squared units.
4	**Skew near zero means bell-shaped?**	→ No — two humps also score zero.

The step we take today

Yesterday a standard deviation of 15 was a fact ABOUT your data. Today it becomes a UNIT you can measure with: “this value is 2 standard deviations above the mean” means the same thing whether you are measuring IQ, rainfall or revenue. That is what makes the rest of Module 2 possible — every test you meet from Day 25 onwards is built on it.

SESSION PLAN

How today's two hours are structured

1 **The bell curve** — 20 min
What it is, and why it keeps appearing

2 **68 – 95 – 99.7** — 20 min
The rule, and what it is good for

3 **Z-scores** — 30 min
The SD as a ruler — comparing unlike things

4 **⚠️ When it does NOT apply** — 15 min
The assumption that quietly breaks

5 **Mini build** — 5 min
A normality check you can trust

6 **Practice notebook** — 30 min
You code, I walk around

Section 4 is in red because assuming normality on data that is not normal is the most expensive mistake in this module.

LEARNING OUTCOMES

By the end of today you will be able to

1

Recognise

Say what makes a distribution normal, and why it recurs

2

Apply

Use 68-95-99.7 to judge whether a value is unusual

3

Standardise

Turn any value into a z-score and compare unlike things

4

Doubt

Check normality before relying on it — and know what breaks

Why this one distribution gets its own session

Not because real data is usually normal — it often is not. Because the AVERAGES of samples are, almost regardless of what the underlying data looks like. That is the Central Limit Theorem, it arrives on Day 27, and it is the reason every hypothesis test in this module works. Today builds the vocabulary for it.

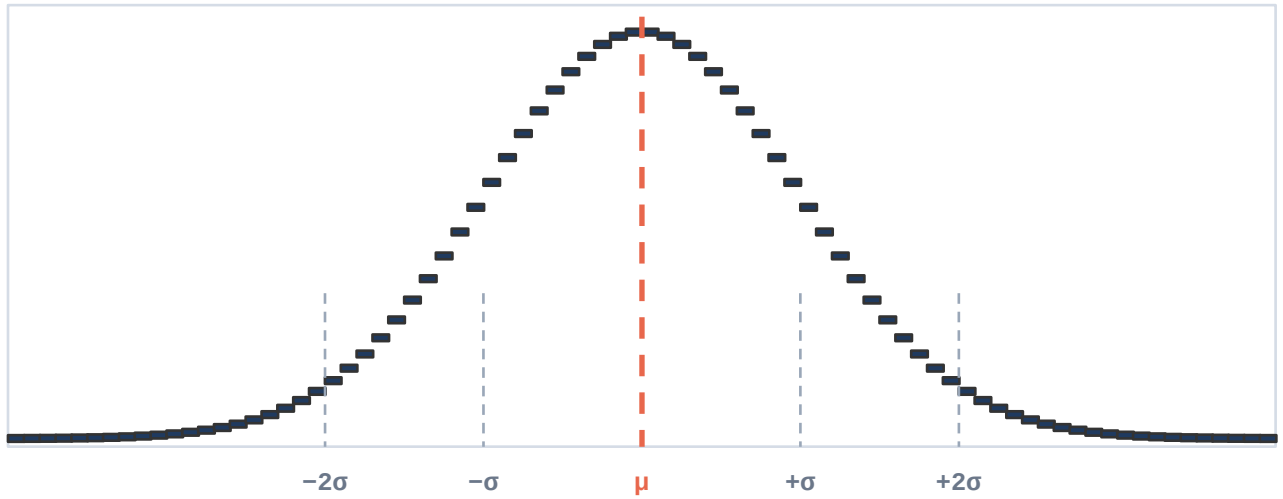
1

The Bell Curve

What it is, and why it recurs · 20 minutes

- Two numbers define it
- Where it comes from
- What it is not

Two numbers define the entire curve



μ sets WHERE it sits **σ sets HOW WIDE it is**

Every normal curve is the same shape

Change μ and the curve slides left or right. Change σ and it gets wider or narrower. Nothing else about it can change.

That is why ONE table of probabilities works for every normal distribution in existence — and why z-scores work at all.

```
stats.norm(loc=μ, scale=σ)
```

Three properties worth naming: it is symmetric (so mean = median = mode), it has a single peak, and the tails go on forever without ever touching zero. That last one matters — a normal distribution always assigns SOME probability to any value, however absurd. Section 4 shows why that bites.

Why this shape keeps turning up

Many small, independent effects adding up

Height is genes plus nutrition plus sleep plus a hundred other things, each nudging you up or down a little. Measurement error is a dozen tiny imprecisions combining.

Whenever an outcome is a SUM of many small independent contributions, the bell curve appears — almost regardless of what each contribution looks like on its own.

⚠️ And when it does NOT appear

When effects MULTIPLY rather than add, you get a long right tail instead — income, city populations, file sizes, word frequencies.

When there is a hard floor at zero and no ceiling, symmetry is impossible. Salaries cannot be negative; waiting times cannot be either.

The real reason it matters — and it is not what you think

The normal distribution is not important because data is usually normal. It often is not. It is important because the AVERAGES of repeated samples are normal almost regardless of the underlying shape — even for income, even for waiting times. That is the Central Limit Theorem, it lands on Day 27, and it is the engine under every test in the rest of this module. Today you are building the vocabulary you will need to state it.

2

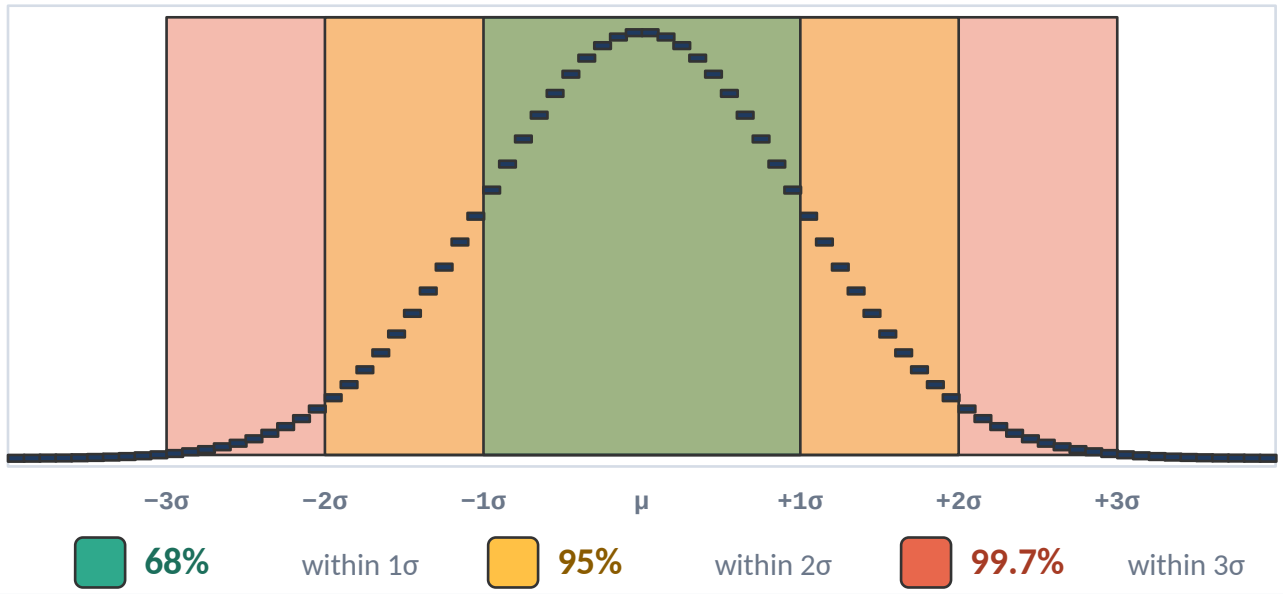
68 – 95 – 99.7

The empirical rule · 20 minutes

- The three bands
- Using it in your head
- The exact figures

Slides by Srinivasa Sai Chava

The one rule worth memorising



The exact figures

Within	Exactly	Outside
1σ	68.27%	1 in 3
2σ	95.45%	1 in 22
3σ	99.73%	1 in 370

The “outside” column is the useful one. A 2σ event happens about once in 22 — unusual, not remarkable. A 3σ event is once in 370.

Use it as a mental ruler. If the mean commute is 32 minutes with an SD of 6, then a 50-minute journey is 3σ out — roughly a once-in-370-days event. You can say that in a meeting without opening a laptop, and it is usually the only calculation anyone needs.

3

Z-scores

The standard deviation as a ruler · 30 minutes

- The formula
- Comparing unlike things
- cdf and ppf
- 1.96

Slides by Srinivasa Sai Chava

One formula, and what it buys you

The formula

$$z = (\text{value} - \mu) / \sigma$$

"How many standard deviations above or below the mean is this?"

```
z = (s - s.mean()) / s.std()
#  the whole column at once — Day 13
```

What standardising does

```
a: mean 100.0  sd 15.00
b: mean   5.0  sd  0.20

after z:  BOTH become mean 0, sd 1
```

Any normal distribution becomes the SAME one. Different units, different scales, same ruler.

The payoff: comparing things that cannot otherwise be compared

Exam	Your mark	Class mean	Class SD	z	
Maths	78	70	8	+1.0	better than ~84% of the class
Physics	65	55	5	+2.0	better than ~98% of the class

The lower raw mark is the better performance. Without z-scores you cannot see that at all.

From a z-score to a probability, and back

cdf — value in, probability out

```
from scipy import stats

#  IQ: mean 100, sd 15
stats.norm.cdf(130, 100, 15)
  ->  0.9772       97.72% score BELOW 130

1 - stats.norm.cdf(130, 100, 15)
  ->  0.0228        2.28% score ABOVE
```

"What fraction is below this value?"

ppf — probability in, value out

```
stats.norm.ppf(0.95, 100, 15)
  ->  124.67
#  the 95th percentile

stats.norm.ppf(0.99, 100, 15)
  ->  134.90
#  the top 1% starts here
```

"What value sits at this percentile?" ppf is the inverse of cdf.

The number you will meet for the rest of this module

```
stats.norm.ppf(0.975)    ->   1.9600
stats.norm.ppf(0.95)     ->   1.6449
```

1.96 is the z-score with 2.5% of the curve beyond it on each side, leaving 95% in the middle. It is where "95% confidence interval" comes from, and you will see it in every formula from Day 29 onwards. It is not a magic number — it is just ppf(0.975).

4

⚠️ When It Does Not Apply

The assumption that quietly breaks · 15 minutes

- Skewed data
- The coincidence trap
- How to check

Apply the rule to salaries and watch it fail

1,000 salaries, skew 4.29

Check	Normal says	Reality	
within 1σ	68.3%	95.0%	badly wrong
within 2σ	95.4%	95.0%	right — by luck
below the mean	50%	93.3%	badly wrong
beyond +3σ	0.13%	4.50%	33x too many
95th percentile	194,667	78,502	off by 116,000

mean − 2σ = −93,758 — a negative salary

⚠️ Look at row 2

The 2σ check PASSES — 95.0% against a predicted 95.5%. If that had been the only thing you tested, you would have concluded the data was normal and moved on.

It is a coincidence. Every other check fails badly.

One passing check is not evidence. Look at the shape.

What this costs you in practice

If you set a “high earner” threshold at the normal 95th percentile you would pick 194,700 and catch almost nobody, when the real cut-off is 78,500. Every downstream decision — a budget, an alert threshold, a sample size — inherits the error. This is the mistake that costs money.

Three ways to check, in order of usefulness

1

Plot it

`A histogram, and a Q-Q plot`

By far the most informative. You see skew, humps and outliers at once.

2

Check skew

`abs(s.skew()) < 0.5`

A quick numeric screen. Over 1 is definitely not normal.

3

A formal test

`stats.shapiro(s)`

p > 0.05 means “no evidence against normal” — which is NOT the same as normal.

⚠️ *A Shapiro test on 50,000 rows will reject normality for a trivial, meaningless deviation. On large data, trust the plot, not the p-value.*

RECAP

Debug corner — eight normal-distribution mistakes

You did	What goes wrong	Fix
assumed normality without looking	every percentile is wrong	plot it first
tested ONE band and stopped	a coincidental pass	check several, and the shape
`stats.norm.cdf(130)`	forgot loc and scale — assumes μ=0, σ=1	`cdf(130, 100, 15)`
used the mean on skewed data	it is not the centre of anything	median and IQR — Day 23
read p > 0.05 as “it is normal”	it means NO EVIDENCE AGAINST	say it that way
Shapiro on 50,000 rows	rejects trivial deviations	trust the plot
z-scores on non-normal data	the percentages do not hold	z still ranks, but does not give probabilities
treated 1.96 as magic	it is just ppf(0.975)	and 1.645 is ppf(0.95)

Row 7 is subtle and worth a moment: a z-score is still a valid ranking on any data. What it stops giving you is a PROBABILITY.

Putting it together — a normality check you can trust

```python
from scipy import stats

def normal_check(s, name="value"):
    """Several independent checks - never just one."""
    s = s.dropna()
    m, sd = s.mean(), s.std()

    bands = {}
    for k in (1, 2, 3):
        obs = s.between(m - k*sd, m + k*sd).mean() * 100
        exp = (stats.norm.cdf(k) - stats.norm.cdf(-k)) * 100
        bands[f"within_{k}sd"] = (round(obs, 1), round(exp, 1))

    out = {
        "n":          len(s),
        "skew":       round(s.skew(), 2),
        "kurtosis":   round(s.kurt(), 2),
        "below_mean": round((s < m).mean() * 100, 1),   # normal: 50
        "pct95_real": round(s.quantile(.95), 1),
        "pct95_norm": round(m + 1.645 * sd, 1),
        **bands,
    }

    #  the verdict uses SEVERAL signals, not one
    bad = (abs(out["skew"]) > 1
           or abs(out["below_mean"] - 50) > 8
           or abs(bands["within_1sd"][0] - 68.3) > 6)
    out["verdict"] = "NOT normal - do not use the rule" if bad else "close enough"
    return pd.Series(out)
```

Three bands, not one
a single pass can be luck

below_mean
asymmetry shows up here first

Both 95th percentiles
the gap is the practical cost

skew AND kurtosis
Day 23's shape numbers

Several signals
any one of them can fire

No p-value
it misleads on large n

Still plot it
no function replaces looking

The salary data passes the 2σ test and fails this one. That is the point of checking more than one thing.

OVER TO YOU

Practice session — 30 minutes

A **Predict the output** — 8 questions

Bands, z-scores, cdf/ppf, 1.96

B **Spot & fix the bug** — 4 questions

Missing loc/scale, p-value misreading

C **Write the code** — 5 tasks

Standardise, compare, percentiles, check

Your notebook

`Day24_Practice_Questions.ipynb`

- Always pass loc and scale
- Plot before you trust a rule
- Raise a hand — I am walking around
- Solutions shared after class

Before using the empirical rule on any column today, check that it applies. Most of them will not.

TODAY IN TWELVE LINES

Recap

1. Two numbers define the whole curve: μ for where, σ for how wide.
2. 68% within 1σ, 95% within 2σ, 99.7% within 3σ.
3. The useful form: outside 2σ is 1 in 22; outside 3σ is 1 in 370.
4. z = (value − μ) / σ — the SD becomes a unit of measurement.
5. Standardising turns ANY normal curve into the same one.
6. A lower raw mark can be the better performance. Only z shows it.
7. cdf: value in, probability out. ppf: probability in, value out.
8. 1.96 is not magic — it is ppf(0.975), leaving 95% in the middle.
9. ⚠️ Real data is often NOT normal. Income, delays, file sizes.
10. On the salary data the 2σ check passed by pure coincidence.
11. So test several things, and look at the shape. One pass proves nothing.
12. p > 0.05 means no evidence AGAINST normal — not that it is normal.

Homework

Run normal_check on every numeric column of your Day 21 project data and say which are normal · Find your own two-exam example and compute both z-scores · Use ppf to find the mark that would put a student in the top 5%

Next class — Topic 2.5

Binomial & Poisson: the distributions for counting things — how many clicks out of 1,000, how many failures this month. Discrete, not continuous.